Prob. 2. (Z&T, Ch. 7, 16.1)

Find <u>all</u> sol-ns to the ↓ Neumann pr-m on the unit circle

(1) $\Delta u = 0, \quad 0 \le r < 1, \quad -\pi \le \theta \le \pi.$

(2) $\frac{\partial u}{\partial n}(1,\theta) = f(\theta) = |\theta| - \frac{\pi}{2}, \quad -\pi \le \theta \le \pi$

Sol-n: First note that for Neumann pr-m to have a sol-n $\int_{\partial\Omega} f(\theta)\,ds = 0$, so

(3) $A_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\,d\theta = \frac{1}{\pi}\int_{-\pi}^{\pi}\left(|\theta| - \frac{\pi}{2}\right)d\theta = \int_{-\pi}^{0}\left(-\theta - \frac{\pi}{2}\right)d\theta + \int_{0}^{\pi}\left(\theta - \frac{\pi}{2}\right)d\theta = \left[-\frac{\theta^2}{2} - \frac{\pi\theta}{2}\right]_{-\pi}^{0} +$

$+ \left[\frac{\theta^2}{2} - \frac{\pi\theta}{2}\right]_{0}^{\pi} = \frac{1}{\pi}\left\{0 - \left(-\frac{\pi^2}{2} - \frac{\pi(-\pi)}{2}\right) + \left[\frac{\pi^2}{2} - \frac{\pi^2}{2}\right]\right\} = 0$ ✓ ⇒ might have a sol-n

because $\int_{\partial\Omega} f\,d\theta \neq 0 \Rightarrow \nexists$ a sol-n.

Then, by the sol-n in class

(4) $u(r,\theta) = \sum_{n=1}^{\infty}\left[\frac{a_n}{n} r^n \cos(n\theta) + \frac{b_n}{n} r^n \sin(n\theta)\right]$ where

(5) $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(n\theta)\,d\theta \qquad \forall n = 1, 2, \ldots$ &

(6) $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\sin(n\theta)\,d\theta \qquad n = 1, 2, \ldots$ Then

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\left(|\theta| - \frac{\pi}{2}\right)\cdot\cos(n\theta)\,d\theta = \frac{1}{\pi}\cdot\left\{\int_{-\pi}^{\pi}\overbrace{|\theta|\cos(n\theta)}^{\text{even}}\,d\theta - \frac{\pi}{2}\cdot\overbrace{\int_{-\pi}^{\pi}\cos(n\theta)\,d\theta}^{=0,\ \text{Pres. HW10.}}\right\}$$

$$= \frac{2}{\pi}\int_{0}^{\pi}|\theta|\cos(n\theta)\,d\theta = \frac{2}{\pi}\cdot\int_{0}^{\pi}\theta\cos(n\theta)\,d\theta = \frac{2}{\pi}\int_{0}^{\pi}\underset{u}{\theta}\,d\underbrace{\left[\frac{\sin(n\theta)}{n}\right]}_{v} =$$

$$= \frac{2}{\pi}\cdot\left\{\theta\cdot\overbrace{\frac{\sin(n\theta)}{n}\Big|_{0}^{\pi}}^{=0} - \frac{1}{n}\int_{0}^{\pi}\sin(n\theta)\,d\theta\right\} = \frac{-2}{\pi n}\left[-\frac{\cos(n\theta)}{n}\right]_{0}^{\pi} =$$

$$= \frac{2}{\pi n^2}\Big[\cos(n\theta)\Big]_{\theta=0}^{\pi} = \frac{2}{\pi n^2}\cdot\Big[\cos(\pi n) - 1\Big]$$

(7) $a_n = \frac{2}{\pi n^2}\Big[\cos(\pi n) - 1\Big]$. $n = 1, 2 \ldots$ similarly

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\Big(|\theta| - \frac{\pi}{2}\Big)\cdot\sin(n\theta)\,d\theta = \frac{1}{\pi}\Big\{\overbrace{\int_{-\pi}^{\pi}|\theta|\cdot\sin(n\theta)\,d\theta}^{\int_{-\pi}^{\pi}\text{odd}=0} - \frac{\pi}{2}\overbrace{\int_{-\pi}^{\pi}\sin(n\theta)\,d\theta}^{\text{odd}=0}\Big\}$$

$$= \frac{1}{\pi}\cdot\Big(-\frac{\pi}{2}\Big)\cdot 0 = 0. \Rightarrow$$

(8) $b_n = 0$ $n = 1, 2 \ldots$ Then (4) becomes

$$u(r,\theta) = \sum_{n=1}^{\infty}\frac{1}{n}\cdot\frac{2[\cos(\pi n)-1]}{\pi n^2}r^n\cos(n\theta) = \sum_{n=1}^{\infty}\frac{2[\cos(\pi n) - 1]}{\pi n^3}r^n\cos(n\theta)$$

(9) $u(r,\theta) = \sum_{n=1}^{\infty}\frac{2[\cos(\pi n) - 1]}{\pi n^3}r^n\cdot\cos(n\theta)$,

Note if $n = 2k$ $\Sigma = 0$, so let $n = 2k-1$, $k = 1, 2 \ldots$

(10) $u(r,\theta) = \sum_{k=1}^{\infty}\frac{-4}{\pi\cdot(2k-1)^3}r^{2k-1}\cdot\cos[(2k-1)\theta]$, is the set of all sol-ns. $\blacksquare$